

Individual differences in reading comprehension in monolingual and bilingual children: The influence of semantic priming during sentence reading

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ABSTRACT

Group differences in semantic priming between young readers with different comprehension levels have been reported, with poor readers showing reduced or no context-independent semantic priming compared to normal readers. However, other studies have not been able to replicate these effects on an individual differences level, even though the spreading of semantic activation is hypothesized to play a role in the reading comprehension process. In the present study, we investigated whether priming during sentence reading, rather than single word priming, could be related to children's reading comprehension scores. A self-paced reading experiment involving both associated and non-associated, context-dependent (functional) and context-independent (categorical) semantic relations was administered to 137 Dutch monolingual and bilingual children. Delayed facilitative priming effects were observed for non-associated context-dependently and context-independently related word pairs, but these were not linked to individual differences in reading comprehension. Monolinguals and bilinguals showed similar performance on almost all language measures, including semantic priming and reading comprehension.

1. Introduction

Reading comprehension is a central skill taught to school-age children, serving as a means of knowledge transfer that becomes increasingly important throughout their school careers and beyond. Nevertheless, the levels of reading comprehension that children attain vary, even setting aside problems related to specific reading difficulties and more general language impairments. Studying the component skills that constitute or support comprehension may help explain these individual differences and serve as a starting point to guide intervention efforts. Vocabulary knowledge is a central element of the comprehension process (cf. Perfetti & Stafura, 2014), which consists of many subcomponents. One of these vocabulary knowledge subcomponents is the interconnectedness of the semantic network, or the structure connecting words and word meanings in the mental lexicon through various types of semantic relations (Aitchison, 2012). High quality lexical knowledge contributes to reading comprehension (Nation & Snowling, 1999; Perfetti, 2007), so it is not only a matter of vocabulary size, but also of the amount and quality of the (semantic) information that becomes available and activated upon word identification. Although this

process of spreading activation toward semantic and situational knowledge is presumed in various models of reading, such as Perfetti's Reading Systems Framework (Perfetti & Stafura, 2014), Kintsch's Construction and Integration model (Kintsch, 1998) and Van den Broek's Landscape model (Van den Broek, Young, Tzeng, & Linderholm, 1999), relatively few studies have been conducted into the role of semantic priming effects in reading comprehension. Does reading about a *forest* automatically activate *tree* and if so, is that beneficial to one's reading comprehension? And to what extent are possible effects mediated by the different kinds of lexical-semantic relations that may exist between the words involved? The present study aims to extend existing research on the connection between comprehension and automatic activation, i.e. semantic priming, in the semantic network, adding to our understanding of the influence of this vocabulary subcomponent on reading comprehension.

In previous research attempting to tease apart the subskills underlying reading comprehension, vocabulary size is identified as an undisputed key player (e.g. Alderson, 2005; Grabe, 2009; Stæhr, 2008; Stanovich, 2000). However, not only the number lexical items known to the reader, but also other components of vocabulary knowledge have a

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role to play, such as depth, fluency and knowledge of semantic relations (e.g. Cremer & Schoonen, 2013; Ouellette, 2006; Swart et al., 2017; Tannenbaum, Torgesen, & Wagner, 2006), and the level of accessibility of this knowledge, i.e. the automatic activation of semantic relations within the semantic network as well (Betjemann & Keenan, 2008; Bonnotte & Casalis, 2010; Nation & Snowling, 1999). In the studies by Bonnotte and Casalis and Nation and Snowling specifically, group differences in semantic priming were found between normal readers and poor readers with reading comprehension delays of at least one year. These studies reported an absence of priming or reduced priming for unassociated category coordinates such as *cow* – *goat* in poor comprehenders. Both words belong to the same superordinate category of animals, and in that sense are semantically related, but are unlikely to elicit one another in a word association task and thus are considered unassociated, i.e., there is no (additional) relation due to contiguity in language use (contrary to, for example, *cat* and *dog*, which have not only a categorical relation, but are also lexically associated). In the case of unassociated categorical relations (*cow* – *goat*), priming effects can only be accounted for by category membership, which is a feature that exists independent of context of word use. Bonnotte and Casalis found that poor readers showed priming for associated functional pairs such as *broom* – *floor*, whereas normal readers did not exhibit such facilitation. The studies by Nation and Snowling (1999) and Bonnotte and Casalis (2010) complement each other in the suggestion that proficient reading comprehension is related to different types of semantic priming in terms of associated or unassociated words, and categorically or functionally related words. However, both studies compared relatively distinct groups of readers. In contrast, recent research attempting to relate individual differences in reading comprehension – rather than group differences – to individual differences in semantic activation has found different results. Both Cremer (2013) and Spätgens and Schoonen (2018) studied normal readers and found no advantages in reading comprehension for children showing more semantic priming of non-associated category relations. It is conceivable that because normal readers are a more homogenous group than those juxtaposed in Nation and Snowling (1999) and Bonnotte and Casalis (2010), individual differences in semantic activation facilitating comprehension may also be more subtle.

A potential way of finding these more subtle effects may be to study priming during reading, rather than single word priming. If advantages in semantic activation would be causally related to reading comprehension skill, a likely mechanism for this contribution would be through activation of existing lexical or conceptual knowledge that may help solve or establish referential or causal connections in a developing mental text model. Such semantic priming during reading comprehension is part of the memory-based processing during reading as postulated by the Landscape model of reading (e.g. Van den Broek et al., 1999; Van den Broek, Rapp, & Kendeou, 2005). These memory-based processes occur as a consequence of the quality of a reader's semantic and background knowledge and tend to happen autonomously, i.e., without the need for strategic attention (Van den Broek et al., 2005). This would suggest that the activation of functionally related or associated words only is less facilitative in establishing text comprehension than activations that also include activation of semantic or categorical relations between words that share features, but may not be associated in other respects. The activation of relevant information helps establishing coherence in sentence and text reading as is described in the Landscape Model (Van den Broek et al., 1999, 2005). The activation of semantic links observed in single-word priming may however not have a close enough correspondence to activation during reading. Correlating comprehension performance to semantic activation during reading may therefore be an additional way to test for the presence of individual differences similar to the group-level effects that have been found in the aforementioned studies by Nation and Snowling (1999) and Bonnotte and Casalis (2010).

The first goal of the present study is therefore to study semantic activation during reading of four types of semantic relations close to those studied by both Nation and Snowling (1999) and Bonnotte and Casalis (2010), but that are described by two dimensions involved: context-dependent versus context-independent semantic relations, and associated versus non-associated. The first contrast distinguishes word pairs related through co-occurrence in context, such as *forest* – *squirrel*, from word pairs related through shared inherent qualities independent of context, such as *cat* – *tiger* (Cremer, 2013; Verhallen & Schoonen, 1993). This distinction is related to the developmental path from episodic to categorical relations observed in vocabulary acquisition (Nelson, 2007), but is considered a continuum, ranging from personal, context-dependent (*squirrel* – *cute*) to taxonomic relations (*squirrel* – *animal*) with levels in-between. The second contrast, association strength, is defined here as the prevalence of a target as a response to its prime in group data from a free word association task. The presence of an associative relation (*cat* – *dog*), which in this definition can occur in addition to various types of semantic relations, is known to cause additional activation compared to non-associated pairs (*cow* – *goat*) in priming studies, the so-called 'associative boost' (see Lucas, 2000, for a comprehensive overview). The distinction between associated and non-associated pairs was crucial in the group differences reported by Nation and Snowling (1999) and Bonnotte and Casalis (2010).

Based on Nation and Snowling (1999) and Bonnotte and Casalis (2010), it appears that especially activation of non-associated, context-independent pairs is predictive of reading comprehension. This notion of the importance of context-independent knowledge is corroborated by evidence from declarative tasks of semantic knowledge, which have shown that children with more context-independent knowledge (Ouellette, 2006) or a better ability to distinguish context-independent relations from context-dependent relations (Cremer & Schoonen, 2013) performed better at reading comprehension tasks. However, to what extent automatic activation of this memory-based knowledge during reading is beneficial to reading comprehension is largely unknown. Given the evidence from off-line tasks and the presumed role of activation of memory-based knowledge in models of reading such as the Landscape Model, we expect to find a positive relation between the automatic activation of non-associated context-independent semantic knowledge and reading comprehension. In addition, by including associated and context-dependently related pairs, we involve a similar contrast to that used by Nation and Snowling (1999) and Bonnotte and Casalis (2010), so that we may be able to identify other effects of semantic activation as well.

A secondary goal of the present study is to compare monolingual and bilingual minority children in these respects. In the Dutch context, about 17% of children in the final year of elementary school are currently from a non-Western immigrant background (CBS, 2016). This minority group is often associated with poorer performance on a range of language measures (cf. CBS, 2014), including knowledge of semantic word relations (Verhallen & Schoonen, 1993). On the other hand, differences between monolinguals and bilingual minority children in the final years of elementary school (age 9–12) appear to be diminishing in large scale educational reports: it may be the case that only bilinguals who do not speak any Dutch at home are at a disadvantage (e.g., compared to bilinguals who use both Dutch and another language at home, cf. Heesters, Van Berkel, Van der Schoot, & Hemker, 2007), or that socio-economic status (SES) is a mediating factor (cf. Kuhlmeier et al., 2014, who controlled for SES differences and found no significant differences between monolinguals and bilinguals on various language measures). Recent studies targeting knowledge of semantic relations and reading comprehension have also found mixed results in the same population: Cremer (2013) demonstrated delays in reading comprehension in bilinguals in two separate studies, and found disadvantages in offline knowledge of context-independent relations and speed of access to semantic word knowledge for bilinguals, but not in semantic

priming. Our previous research found no significant differences in terms of semantic activation or reading comprehension (Spätgens & Schoonen, 2018). In the present study, we will address the difference between these groups again to determine whether our previous findings are confirmed in another sample from the same population and with another task to assess semantic priming.

1.1. Description of the current research

In order to study semantic activation during reading, a single sentence self-paced reading experiment was designed. The sentences contained prime-target word pairs that bore a semantic relation and these relations could be described as associated or non-associated and context-dependent or context-independent. Facilitation on each of these four semantic types was measured by contrasting sentences including related pairs with the same sentences in which the targets were preceded by unrelated primes. Individual facilitation effects emerging in this task were then used to predict reading comprehension scores, while controlling for vocabulary size, declarative knowledge of context-independent semantic relations and word decoding speed. The target population consists of children at the end of primary education and who are of mixed socio-economic and L1 background, but who all received their education in the dominant language, Dutch. To the best of our knowledge, this is the first study attempting to relate semantic priming during reading to individual differences in reading comprehension, and thus aims to fill a gap in the current knowledge.

2. Method

2.1. Participants

Participants were students of fifth grade in primary education and were recruited through their schools, which were selected based on location, that is the urban western part of the Netherlands. All seven schools were in neighborhoods with an overall average to low SES (SCP, 2015) and mixed populations of monolingual residents and multilingual residents with a migration background. However, all participating children had received their education in the Netherlands, and were excluded if they had not received Dutch education from grade 1 onwards. The regular education offered by the schools was in Dutch only. All parents were informed via the schools on the contents and purpose of the study through a passive informed consent procedure approved by the ethics committee of the Faculty of Humanities of the University of Amsterdam. Thirteen children from the recruited groups did not participate in the tasks because their parents objected.

In total, 160 children took part.¹ Of this group, the data from 23 children were removed for either developmental reasons such as being diagnosed with dyslexia (nine children), ADHD (one child), or general language delays (three children); due to incomplete data or equipment failure (six children); because of low accuracy on the self-paced reading task (57%, one child); or because the participants had started elementary school in the Netherlands after grade 1 (three children).

The final sample of 137 children had a mean age of 11;2 (*SD* = 0;5) and included 61 monolingual and 76 bilingual participants, 48.9% girls. The participant descriptives are provided in Table 1. All but 15 children were born in the Netherlands. Fourteen of these fifteen children arrived in the Netherlands before or at the age of Kindergarten education, one child arrived at the start of ‘formal’ education (Grade 1). The monolingual children reported that Dutch was their first language and the language they spoke at home. A small minority of the monolingual children mentioned that they are incidentally exposed to another

Table 1
Participant descriptives by language group.

	Total N	Girls	Boys	Age (sd)
Monolingual	61	28	33	11;1 (0;5)
Bilingual	76	39	37	11;2 (0;5)
Total	137	67	70	11;2 (0;5)

language than Dutch, because of relatives or parents speaking another language among themselves. However, these children do not speak the language themselves. The bilingual children reported to speak another language than Dutch as their first language; they can be considered sequentially bilingual. Sometimes they also speak Dutch at home, in addition to the first language. The other languages constitute a rather heterogeneous set of languages, ranging from Berber (6), Kurdish (4), Turkish (10) and Surinamese (7) to Ghanese (4), Spanish (6), English (7) and various multilingual combinations of languages. The mono-bilingual division is somewhat arbitrary in the sense that it consists of a continuum from monolingual Dutch to bilingual or multilingual, non-native Dutch.

2.2. Materials

In this section, we will describe the experimental self-paced reading task that was used to collect semantic priming data extensively (Section 2.2.1), as this was not a standard task. The other language tasks for reading comprehension, vocabulary size, context-independent semantic knowledge and word decoding, Sections 2.2.2–2.2.5 respectively, and the language interview (Section 2.2.6) are more briefly introduced as they are more standardized measurements.

2.2.1. Self-paced reading task

A single sentence self-paced reading task with a non-cumulative moving window paradigm was used (cf. Just, Carpenter, & Woolley, 1982). In this paradigm, an initially fully masked sentence is presented, after which the first button press reveals the first word. Following a second button press, the first word is re-masked and the second word is revealed, etc. An example sentence of three words would be presented in the following four stages, each presented on a new screen after a button press:

--- ----- .
Mia ----- .
--- loves ----- .
--- ----- words.

In this way, reading times for each word can be measured. To ensure participants read the sentences for comprehension, a yes/no plausibility judgement was required after each sentence. To keep the participants focused, 50% of stimulus sentences was nonsensical. Participants were expected to score (close to) 100% correct in their judgements (see below).

2.2.1.1. Word pair construction. To test for semantic priming, 56 prime-target word pairs were created to be included in the test sentences. Four types of semantic relations in a 2 × 2 contrast were used: prime-target word pairs in the sentences were either context-dependently or context-independently related, and the paired words are also associated or not (non-associated). Example word pairs of each type are provided in Table 2.

As becomes clear from the examples, context-independent pairs are represented by category relations, either subordinate – superordinate

¹ In addition, two groups participated in two separate pilot sessions following the same passive informed consent procedure, testing earlier versions of the self-paced reading task (36 children in total).

Table 2
Example related word pairs by semantic type.

	Context-independent		Context-dependent	
Non-associated	cat	tiger	desert	snake
	trumpet	flute	captain	harbor
Associated	banana	fruit	hunter	rifle
	ocean	sea	store	money

pairs, or coordinate pairs. Context-dependent pairs were either *location – animal/person often found at this location*, or *person – location/object this person often occurs at/with*. The same types of relations were used in Spätgens and Schoonen (2018), although subordinate – superordinate pairs were added to be able to create enough associated, context-independently related pairs. Primes and targets were selected from the 5000 most frequent words in the Dutch school language corpus from Schrooten and Vermeer (1994).

All non-associated pairs used in this experiment are from a pool of potential stimuli developed for the semantic priming experiment in Spätgens and Schoonen (2018). These pairs are controlled for relatedness through a questionnaire in which items were rated by 33 adult speakers of Dutch, so that context-independent and context-dependent pairs were equally strongly related. Association strength for these items was determined based on a previous study in the same target population (Spätgens and Schoonen, submitted). For this data base, three association responses to the primes were elicited from at least 50 children to determine association strength with potential target words. Non-associated targets in our test items were never provided as first responses, and at most once as second or third responses. In contrast, the associatively related pairs were selected based on their regular occurrence as association responses in this data base. Across the two sets, the associated targets appeared 6.7 times as first responses, 3.8 times as second responses and 3.0 times as third responses, on average.

All translated word pairs are provided in Appendix A, and length and frequency information for primes and targets by semantic category is provided in Appendix B. Related and control primes were matched for frequency and length within each of the semantic types.

2.2.1.2. Sentence construction. The related prime-target pairs were embedded in sentences, each paired with another sentence which was identical except for the replacement of the related prime by the semantically unrelated prime. The construction of the test sentences was done according to a number of criteria. Firstly and most importantly, both the related and unrelated sentences were required to be equally acceptable.

Secondly, various restrictions were placed on the positioning of the prime-target pairs in the sentence. Prime words were always preceded by at least one word to be sure the participant is ready for the task. Prime-target pairs were separated by at least one and at most three words, which did not steer the meaning of the sentence toward the target. Finally, targets were followed by at least five words of which the first three functioned as potential spillover positions, that is to allow for the measurement of delayed priming effects. The faster or easier processing of the target word may (also) show in the reading times of the following words (Rayner, Sereno, Morris, Schmauder, & Clifton Jr., 1989), hence the need for a spillover region. The spillover region was followed by at least two words to allow for so-called wrap-up effects, as it is known from psycholinguistic research that the processing of the final words of a sentence or clause may take up extra time that may interfere with our reaction time measurements (Just et al., 1982; Rayner et al., 1989).

Thirdly, both plausible and implausible sentences were designed to be plausible up until the fourth word after the target, so that slowed reading due to an implausible turn of events would not be captured in the spillover region (target word + 3 words). Overall, sentence length

was kept similar across plausible and implausible sentences ($M = 12.8$ words, $SD = 1.53$ and $M = 12.6$ words, $SD = 1.49$, respectively) so that length could not serve as a clue for the plausibility judgement. Plausible and implausible translated examples are provided below.

Plausible example (context-dependent, unassociated prime-target pair).

The old	captain	looks at the	harbor	<i>while the big</i>	boat sails away.
The old	lady	looks at the	harbor	<i>while the big</i>	boat sails away.
	<u>Prime</u>		<u>Target</u>	<u>Spillover region</u>	

Implausible example (context-independent, unassociated prime-target pair).

Dina draws a	cat	next to the	tiger	<i>on the small</i>	soft chewing gum.
Dina draws a	sun	next to the	tiger	<i>on the small</i>	soft chewing gum.
	<u>Prime</u>		<u>Target</u>	<u>Spillover region</u>	

In all, 112 test items were constructed, 56 related and 56 unrelated sentences. These were divided equally across the four semantic types (14 pairs i.e. 28 sentences for each type) and within semantic types, half was plausible and half was implausible. All test sentences (in Dutch) are provided in Appendix C.

2.2.1.3. List compilation. To compile the lists with stimulus sentences, the full set of 112 test sentences was divided in two halves, each including seven related and seven unrelated items from each of the four semantic types, with equal numbers of plausible and implausible items. Each list thus contained only one sentence of each of the pairs, meaning that within children, the halved subsets within each semantic type were used as related and control items.² Additionally, seventeen similarly structured plausible and implausible dummy sentences were used as practice trials and start-up trials at the start of the experiment and after the half-way break. In total, each participant therefore read and responded to 73 items, of which 38 were plausible and 35 were implausible. The startup trials contained a few more plausible items to create an easier start to the experiment and after the break.

Of each of the two lists, three pseudo-randomized versions were created with different orderings. Within each of these six versions, related and unrelated sentences of each of the four semantic types were spread evenly across the list, occurring roughly equally often in earlier and later trials both before and after the break. No more than four consecutive items were of the same plausibility type.

2.2.1.4. Presentation. The experiment was run on two identical laptops using E-Prime software (Schneider, Eschman, & Zuccolotto, 2002). Sentences were presented in a monospaced font (Courier New), and sized to fit a single line on the test computers. Each trial began with a white screen for 500 ms, followed by the first sentence screen with a fully masked sentence. Masking was done by replacing all letters and commas with hyphens (–), leaving punctuation at the end of the sentences unchanged.

Participants pressed the space bar to reveal each individual word in turn. If a sentence contained a comma, the comma was revealed and re-masked together with the preceding word. After the last word in the sentence was revealed, a final space bar press led to a screen with a green V and a red X for the plausibility judgement to be made. The ‘C’

² In the first pilot group, a design with two blocks was tested in which each child was administered both versions, with the order counterbalanced across children. However, in this setup the experiment became too long for the children to remain fully concentrated, and in addition all children reported recognizing the sentences in the second block, making the measurement invalid.

and 'M' keys were marked with a green and red sticker respectively, to serve as the plausible and implausible answer options. After the answer was provided, a white screen appeared again for 500 ms, automatically followed by the next masked sentence. Reading times per word, i.e., between space bar presses, and the speed and accuracy of the plausibility judgement were recorded, the latter as a check on cooperative test behavior.

2.2.2. Reading comprehension

Reading comprehension was tested using the same shortened version of *Begrijpend Lezen 678* [Reading Comprehension grades 4 5 6] (Aarnoutse & Kapinga, 2006) used in Spätgens and Schoonen (2018) and in Cremer (2013, chapter 5). Both superficial and in-depth comprehension is tested in 32 questions on five short texts. The test is preceded by an example text with one multiple choice question with four answer options, and three true/false statements. The internal consistency of the task in this sample was satisfactory (Cronbach's $\alpha = 0.73$).

2.2.3. Vocabulary size

A standardized vocabulary size test was used, the Cito *Leeswoordenschat* [Reading vocabulary] test by Verhoeven and Vermeer (1995). In this test, target words are provided in a neutral sentence and participants are required to select the correct definition from four answer options. There are two practice items and 32 test items. Internal consistency for the vocabulary task was somewhat lower than for reading comprehension and the context-independent semantic knowledge task (below), but not unsatisfactory: Cronbach's $\alpha = 0.62$.

2.2.4. Context-independent semantic knowledge

Declarative context-independent semantic knowledge was assessed using the Word Associates Test (WAT, Schoonen & Verhallen, 2008). This task consists of 30 test items plus two practice items, each including one target word and six answer options. Participants are required to select three answer options that are context-independently related to the target word (e.g., *banana – fruit*). Distractor items include unrelated and context-dependently related words (e.g., *banana – car* and *tasty*, respectively). Items were scored correct, i.e. awarded one point, if all three context-independently related words were selected. Any other answer pattern received zero points. Two differently ordered versions were randomly distributed in the class test setting. Internal consistency in this sample was satisfactory with Cronbach's $\alpha = 0.71$.

2.2.5. Word decoding

Word decoding speed was measured by using the two most difficult lists from the standardized *Drie Minuten Toets* [Three Minutes Test] (Verhoeven, 1992). For each list, participants were required to read as many words as possible out loud within 1 min. This results in two scores per participant: the number of words read minus the number of errors made on each respective list. As an approximation of the internal consistency, we computed the correlation between the scores on the two lists, which was $r = 0.91$. We then averaged the two list scores to create one score per participant.

2.2.6. Language interview

The participants' language status was determined by means of a short language interview including questions on their home language (s), with whom the languages are spoken and the age at which they started school in the Netherlands.

2.3. Procedure

Participants performed the tasks in the school setting, which required one to two school days per class. All tasks were administered by the first author or one of two trained test assistants. Testing started in class with reading comprehension, vocabulary size and the WAT.

Afterwards, the children performed the self-paced reading experiment individually in a quiet room, followed by the word decoding task and the language interview.

As was discussed in the Materials section, each of the classroom-administered tasks contained a few example questions, which were discussed according to the protocols prescribed by the respective instruction manuals. Classroom testing lasted 90 min in total.

The individual sessions started with the self-paced reading task. Participants were explained the self-paced reading procedure, and told to read as quickly and accurately as possible. The plausibility judgement was defined as a decision on whether sentences were 'logical' and 'made sense'. The experimenter provided a few example sentences similar to the experimental items for the child to judge verbally and discussed the answers. A short written reinforcement of the instructions was provided at the start of the experiment, followed by the practice items, for which feedback was provided on-screen. After the practice trials, the experimenter asked whether everything was clear and again emphasized the importance of reading as quickly and as accurately as possible. No further feedback was provided during the experiment.

Next, the two word lists for word decoding were administered, again according to the protocols set by the task's instruction manual, followed by the language interview. Individual sessions lasted about 20–30 min.

2.4. Data handling and analysis

All analyses were performed in R version 3.4.1 (R Core Team, 2017). The reading time data and the reading comprehension scores specifically were modelled using the lme4 package for mixed effects models (Bates, Mächler, Bolker, & Walker, 2015).

For the reading time analyses, all dummy items were removed before the analyses. The high mean accuracy on the plausibility judgement task (90% correct) indicated that participants performed the task by indeed reading for comprehension and performed the task as was intended. None of the participants in the final data set had an accuracy score lower than 70%. Because the sentences were designed to become implausible after the spillover region, i.e. the region of our reading time measurements, and accuracy was not predictive of reading times in any of the four target sets (all p 's > 0.05), both correctly and incorrectly answered items were included in the reading time analyses. Three sentences had comparatively low average accuracy, between 55% and 60% (all other items were > 70% accurate), but were not outliers in terms of reading times on the targets and spillover words.³

For reasons of brevity, the set of targets itself will be referred to as T0, and the following words in the spillover region will be referred to as S1 (target +1), S2 (target +2) and S3 (target +3). Minimal data cleaning was performed on the reading times on each of these target positions: reading times shorter than 100 ms were removed, and personal outlier scores were removed by trimming the data that were ± 2.5 participant SD from the participant mean for each position.

A strict protocol was used to identify the effects of relatedness for each of the semantic types in the targets and spillover regions. In the first step, reading times for the target and each spillover position (T0 to S3), transformed using the natural log to normalize score distributions, were modelled separately using mixed effects models, which included the following covariates as fixed effects:

- The frequency (log transformed using the natural log) and length (in letters) of the target words – i.e. T0, S1, S2 or S3.
- The position of the sentence in which the target occurred in the experiment, taking into account that reading times speeded up

³ We also checked that for all reported models, running on a trimmed dataset involving only reading times from correctly answered items led to the same conclusions, which was the case.

consistently throughout the experiment (cf. Hofmeister, 2011).

- The reading time of the prime (log transformed using the natural log), controlling for difficulties reading the prime that may affect reading times on the targets (cf. Baayen, Davidson, & Bates, 2008).
- The participant's word decoding speed score, to control for individual differences in reading speed which may affect the shape of the priming effect (cf. Kliegl, Masson, & Richter, 2010).

The rest of the fixed part of the model consisted of the four-way interaction and all lower interactions between the variables of interest Language Group, Condition, Context-Dependency and Association, which were all contrast coded and centered binary factors. The four-way interaction was thus: Language Group [-0.5 monolingual; $+0.5$ bilingual] $\times$ Condition [-0.5 related; $+0.5$ unrelated] $\times$ Context-Dependency [-0.5 context-independent; $+0.5$ context-dependent] $\times$ Association [-0.5 non-associated; $+0.5$ associated]. This interaction simultaneously takes into account the influence of Condition on the 2×2 semantic contrast and differences between the language groups.⁴

Finally, the random part of the models included random intercepts for the participant, target and school levels. Random slopes for the variables of interest and their interactions were included as much as possible, following Barr, Levy, Scheepers, and Tily (2013). This meant that the three-way and lower interactions between Condition $\times$ Context-dependency $\times$ Association were included as random slopes for participants, and where possible, a random slope for Language Group by targets was also estimated. However, to attain convergence the latter slope needed to be omitted for some models.

To determine effects of Condition within the two semantic contrasts, these main models were examined for significant interactions involving the Condition factor and the Context-Dependency and/or Association factors. If a significant interaction was found, it was examined by breaking up the target set first by Context-Dependency or Association, depending on the interaction that was found to be significant. A separate model was then fitted for each subset with the same characteristics as described above, except that now a three-way interaction was the largest interaction involved (i.e. Language Group $\times$ Condition $\times$ Association if the data set were split by Context-Dependency). If an interaction between Condition and, in this example, Association was found to be significant in this step, the relevant data set was again split up and analyzed in the same way.

If a significant main effect of Condition without interactions was observed at any of these steps, a semantic priming score was calculated for the relevant semantic set. First, the log transformed reading times were residualized using a model including all control variables and random intercepts for items, participants and schools. The priming scores were then calculated based on these residualized reading times for each participant by subtracting the individual mean for related items from the individual mean for unrelated items.

3. Results

3.1. Descriptives

The scores for the two language groups on the standardized tasks measuring reading comprehension, vocabulary size, context-independent semantic knowledge and word decoding speed are presented in Table 3. All differences are in the expected direction, with monolinguals showing slightly higher scores. However, independent samples

⁴ Because the number of words between primes and targets varied across sentences, we checked whether model fit was improved by including another factor representing this difference in the interaction, which was not the case. For most models there was no significant difference in deviance, and in all cases, deviance for the simpler model excluding the effect of distance between prime and target was lower.

t-tests showed that there were no significant differences for reading comprehension, vocabulary size and word decoding (all *ps* > 0.05), and the effect sizes for these tasks were negligible. Only for context-independent semantic knowledge a small effect (Cohen, 1969) appears: Cohen's *d* = 0.36. This difference was also statistically significant: *t* (133.22) = 2.1, *p* = 0.038.

3.2. Self-paced reading task

As discussed in the Method section, mean accuracy on the plausibility judgement task was 90%, which is taken as an indication of good participant cooperation, and both correct and incorrect items were kept in the dataset for the analyses (see Procedure). Mean reading times at the target and spillover positions are provided in Table 4, and a more detailed visualization of the reading times per language group and semantic set at each position is presented in Fig. 1.

Following the protocol described in the Method section, reading times of words at all four positions, target and spillover, were examined systematically, starting with models including all four semantic sets and the full four-way interactions. The model estimates for the fixed effects of the variables of interest (Language Group, Condition, Context-Dependency and Association) are provided in Table 5 for each of the four full models.⁵

Only reading times for the spillover positions at S1 and S2 were found to show significant interactions involving Condition, which will be discussed in more detail below. No significant interaction involving Condition was found at T0 and S3, suggesting that there were no significant differences between the related and unrelated condition at these positions. A significant interaction between Context-Dependency and Association does appear at S3, which suggests that the words at this spillover position were not comparable across the four semantic sets, independent of Condition. Apparently, there were inherent differences between the semantic sets in terms of the difficulty of reading these words, which persisted even after controlling for length and frequency. The significant main effects of language group at T0, S1 and S3 and to a lesser extent the marginal effect at S2 (*p* = 0.058) suggest that bilingual participants read the targets and spillover words faster. The significant interaction between Language Group, Context-Dependency and Association at S2 suggests that there is a difference in reading speed between the language groups, which depends on the semantic types. The exact shape of this interaction will however be dependent on the other significant interactions involving Context-Dependency and Association, which will be discussed further below.

3.2.1. Priming at S1

For reading times at the S1 position, the interaction between Condition and Association was significant in the full model. The S1 reading times were therefore split into two datasets, along the Association dimension. In the associated set, no significant effects of the variables of interest or their interactions was found, except for a significant main effect of Language Group (β = -0.061 (0.031), *t* = 2.0, *p* = 0.046), which is consistent with the significant main effect in the full S1 model. No significant effect of Condition was thus observed for the associated items. For the non-associated set, again, a main effect of Language Group was found, with faster reading times for the bilingual group. In addition, a main effect of Condition with longer reading times for the unrelated items indicated a delayed facilitation effect of non-associatively related targets, whether context-independently or context-dependently related. The β estimate of 0.015 in log-transformed reading time translates to a facilitative priming effect of 1.5% of the mean reading times in this set (450.6), or about 6.8 ms, which is rather

⁵ Note that fixed effects for control variables and the random structure are omitted for reasons of brevity and clarity in all tables reporting estimates for the priming models.

Table 3Descriptives for standardized tasks by language group, and Cohen's *d* for difference between means.

	Total (N = 137)		Monolingual (N = 61)		Bilingual (N = 76)		<i>d</i>
	M	SD	M	SD	M	SD	
Reading comprehension	23.18	4.39	23.44	4.38	22.96	4.41	0.11
Vocabulary size	16.12	4.14	16.33	4.01	15.95	4.27	0.09
Context-independent semantic knowledge	19.21	4.37	20.07	4.06	18.53	4.51	0.36*
Word decoding	93.05	16.08	93.70	14.81	92.53	17.12	0.07

* Statistically significant difference, $p = 0.038$.**Table 4**

Mean raw reading times at the target and spillover positions.

	M	SD
T0 (N = 7463)	454.5	171.4
S1 (N = 7459)	453.1	158.3
S2 (N = 7450)	443.0	151.7
S3 (N = 7444)	446.5	152.1

N = number of observations (number of participants times number of items).

small. The model estimates are presented in Table 6.

3.2.2. Priming at S2

The full model of the reading times at S2 showed significant interactions between Condition and Context-Dependency, and between Condition, Context-Dependency and Association and the S2 reading times were therefore firstly broken up along the Context-Dependency dimension. No significant effects of the variables of interest or their interactions were observed for the context-dependent items. The

context-independent items however did show a significant interaction. The estimates for the priming model of the S2 context-independent items are provided in Table 7.

A main effect of Language Group was observed again, combined with a significant interaction between Language Group and Association. Furthermore, a significant main effect of Condition was found, but the significant interaction between Condition and Association suggests that this effect was dependent on the type of association. Both interactions were studied in more detail by splitting the data set one more time along the Association dimension. In the resulting associated, context-independent set no significant effects of the variables of interest were found. In contrast, in the non-associated, context-independent set, significant main effects did arise, which clarify the shape of the higher-order interactions in the previous model. Table 8 presents the estimates for this latter set. The significant main effects of both Language Group and Condition in absence of an interaction indicate that the bilinguals read words in this set faster than the monolinguals, and that there was a significant facilitation effect for words at S2 following non-associated, context-independently related targets. The β estimate of 0.033 for the

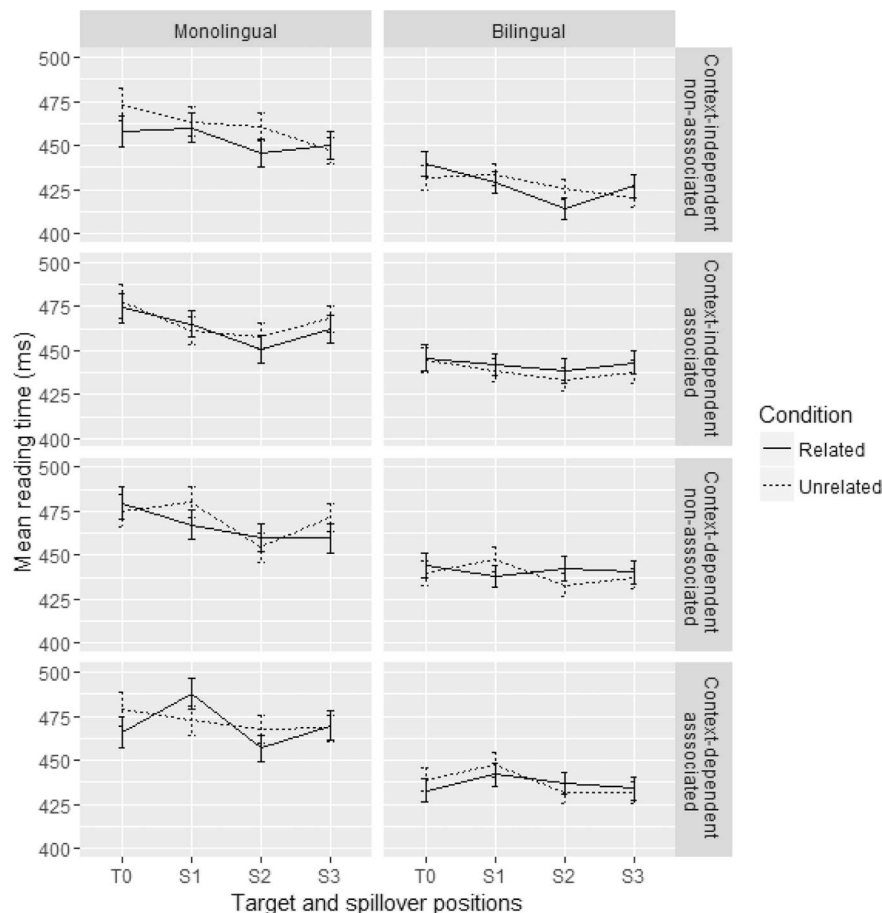
**Fig. 1.** Raw reading times on targets and spillover words by semantic type, condition, and language group.

Table 5
Fixed effects of variables of interest for full models of log transformed reading times at the T0, S1, S2 and S3 positions. N = number of observations.

	T0 (N = 7463)			S1 (N = 7459)			S2 (N = 7450)			S3 (N = 7444)		
	Estimate (se)	t	p	Estimate (se)	t	p	Estimate (se)	t	p	Estimate (se)	t	p
Language group	-0.068 (0.001)	-2.0 (0.035)	0.050 [*]	-0.062 (0.003)	-2.0 (0.031)	0.046 [*]	-0.059 (0.006)	-1.9 (0.031)	0.058	-0.069 (0.001)	-2.1 (0.032)	0.032 [*]
Condition	-0.005 (0.006)	-0.7 (0.006)	0.466	0.012 (0.002)	0.5 (0.005)	0.613	0.006 (0.007)	1.3 (0.005)	0.184	0.016 (0.009)	0.2 (0.005)	0.855
Context-dependency	-0.010 (0.007)	-1.5 (0.007)	0.129	0.002 (0.010)	1.1 (0.011)	0.254	0.003 (0.010)	0.7 (0.011)	0.485	0.006 (0.009)	0.7 (0.009)	0.518
Association	-0.008 (0.011)	-0.7 (0.011)	0.482	0.014 (0.010)	1.4 (0.010)	0.172	-0.013 (0.010)	-1.4 (0.010)	0.172	-0.014 (0.010)	-1.4 (0.010)	0.167
Language group × condition	-0.008 (0.012)	-0.7 (0.012)	0.466	-0.014 (0.011)	-1.3 (0.011)	0.189	0.008 (0.011)	0.7 (0.011)	0.466	-0.010 (0.010)	-1.0 (0.010)	0.344
Language group × context-dependency	0.006 (0.011)	0.6 (0.011)	0.582	0.008 (0.010)	0.8 (0.010)	0.427	-0.023 (0.010)	-2.3 (0.010)	0.023 [*]	0.010 (0.010)	1.0 (0.010)	0.342
Condition × context-dependency	0.000 (0.012)	0.0 (0.012)	0.997	-0.001 (0.010)	-0.1 (0.010)	0.889	0.005 (0.010)	0.5 (0.010)	0.625	-0.007 (0.010)	-0.7 (0.010)	0.472
Language group × association	0.005 (0.011)	0.5 (0.011)	0.601	-0.025 (0.010)	-2.4 (0.021)	0.016 [*]	-0.002 (0.010)	-0.2 (0.010)	0.865	-0.003 (0.010)	-0.3 (0.010)	0.784
Context-dependency × association	-0.013 (0.013)	-1.0 (0.022)	0.296	-0.007 (0.021)	-0.4 (0.020)	0.722	-0.021 (0.020)	-1.0 (0.020)	0.301	-0.037 (0.021)	-2.1 (0.018)	0.034 [*]
Language group × condition × context-dependency	0.017 (0.022)	0.8 (0.022)	0.428	0.024 (0.020)	1.2 (0.021)	0.231	-0.016 (0.020)	-0.8 (0.020)	0.422	-0.002 (0.020)	-0.1 (0.020)	0.934
Language group × condition × association	0.031 (0.021)	1.5 (0.021)	0.141	0.026 (0.021)	1.3 (0.021)	0.209	-0.009 (0.019)	-0.5 (0.019)	0.642	0.006 (0.020)	0.3 (0.020)	0.773
Language group × context-dependency × association	0.001 (0.023)	0.1 (0.023)	0.956	-0.016 (0.020)	-0.8 (0.020)	0.428	-0.044 (0.019)	-2.3 (0.019)	0.021 [*]	-0.020 (0.020)	-1.0 (0.020)	0.319
Condition × context-dependency × association	0.035 (0.022)	1.6 (0.022)	0.107	0.003 (0.020)	0.1 (0.020)	0.896	0.053 (0.020)	2.7 (0.020)	0.008 ^{**}	-0.012 (0.020)	-0.6 (0.020)	0.568
Language group × condition × context-dependency × association	-0.061 (0.044)	-1.4 (0.044)	0.164	0.051 (0.040)	1.3 (0.040)	0.208	-0.005 (0.040)	-0.1 (0.040)	0.907	0.050 (0.041)	1.2 (0.041)	0.223

* $p < 0.05$.** $p < 0.01$.**Table 6**Fixed effects of variables of interest for log-transformed reading times at S1 for the non-associated set ($N = 3728$ observations).

	Estimate (se)	t	p
Language group	-0.069 (0.031)	-2.2	0.027 [*]
Condition	0.015 (0.007)	2.1	0.035 [*]
Context-dependency	0.014 (0.014)	1.0	0.330
Language group × condition	0.002 (0.014)	0.1	0.896
Language group × context-dependency	-0.005 (0.016)	-0.3	0.746
Condition × context-dependency	0.007 (0.014)	0.5	0.638
Language group × condition × context-dependency	-0.001 (0.029)	0.0	0.971

* $p < 0.05$.**Table 7**Fixed effects of variables of interest for log-transformed reading times at S2 for the context-independent set ($N = 3731$ observations).

	Estimate (se)	t	p
Language group	-0.062 (0.031)	-2.0	0.044 [*]
Condition	0.018 (0.007)	2.7	0.008 ^{**}
Association	0.013 (0.017)	0.7	0.462
Language group × condition	-0.005 (0.014)	-0.4	0.729
Language group × association	0.027 (0.014)	1.9	0.047 [*]
Condition × association	-0.028 (0.013)	-2.1	0.033 [*]
Language group × condition × association	-0.006 (0.027)	-0.2	0.813

* $p < 0.05$.** $p < 0.01$.**Table 8**Fixed effects of variables of interest for log-transformed reading times at S2 for the non-associated, context-independent set ($N = 1873$ observations).

	Estimate (se)	t	p
Language group	-0.069 (0.029)	-2.4	0.018 [*]
Condition	0.033 (0.009)	3.5	0.000 ^{***}
Language group × condition	0.000 (0.019)	0.0	0.999

* $p < 0.05$.*** $p < 0.001$.

log-transformed reading time translates to a facilitative priming effect of 3.3% of the mean reading times in this set (434.7), or about 14.3 milliseconds. This effect holds for both language groups, given the absence of an interaction between Language Group and Condition.

In sum, the analyses of the self-paced reading task show a few priming effects, located in the spillover region, that is at one and two positions after the target word. The priming occurs for semantically related words that are not associatively related, and at position S2 it is limited to the context-independently related words. The next question to address is whether these semantic priming effects are related to readers' reading comprehension.

3.3. Semantic priming and reading

In the second step, semantic priming scores based on residualized reading times were calculated for the non-associated S1 targets and the non-associated, context-independent S2 targets. The descriptives by language group are provided in Table 9. As was also clear from the previous analyses, there are no significant differences between the language groups (both p 's > 0.05).

The two individual priming scores were entered in two separate mixed effects models as predictors of the reading comprehension scores, together with language group and their interaction, and the control variables vocabulary size, WAT and decoding. A random effect controlling for school variance was included. The fixed effects of these two models are provided in Table 10 (non-associated priming at S1) and

Table 9

Descriptives for priming scores based on residualized reading times by language group (N = number of participants).

	Total (N = 137)		Monolingual (N = 61)		Bilingual (N = 76)		
	M	SD	M	SD	M	SD	d
S1 non-associated priming	0.01481	0.07425	0.01360	0.07810	0.01578	0.07152	0.03
S2 non-associated, context-independent priming	0.03267	0.09978	0.03280	0.10663	0.03256	0.09464	0.00

Table 10

Fixed effects estimates for predictors of reading comprehension scores, including non-associated priming at the S1 position (N = 137).

	Estimate (se)	t	p
Intercept	11.22 (2.20)	5.1	0.000***
Vocabulary size	0.33 (0.09)	3.7	0.000***
Context-independent semantic knowledge	0.28 (0.09)	3.3	0.001**
Word decoding	0.01 (0.02)	0.5	0.625
S1 non-associated priming	8.34 (4.43)	1.9	0.060
Language group	0.08 (0.68)	0.1	0.901
S1 non-associated priming × language group	−0.68 (8.88)	−0.1	0.939

** $p < 0.01$.*** $p < 0.001$.

Table 11 (non-associated context-independent priming at S2). As expected, vocabulary size and context-independent semantic knowledge were significant predictors of the reading comprehension scores in both models, while Language Group did not make a significant contribution. Neither the S1 nor the S2 priming scores exert a significant influence on the comprehension scores or show a significant interaction with Language Group, but the main effect of S1 was marginal. However, inclusion of neither the S1 nor the S2 priming scores represented a significant improvement in model fit, i.e., reduction of the deviance index, compared to a base model including only the control variables.⁶

4. Discussion and conclusions

First, we will discuss the semantic priming effects we found with our sentence reading task in the spillover regions S1 and S2, and the expected and possible effects we did not find for the target words (T0) and in the more distant spillover region S3, respectively (Section 4.1). The relationship of these priming effects with reading proficiency is discussed next (Section 4.2). It turned out that language background did not play an essential role in the results (Section 4.3).

4.1. Semantic priming during sentence reading

We observed no significant priming effects for the target word (T0), but we did find two delayed significant effects in the spillover region (S1 and S2, respectively). These effects concerned non-associated word pairs. No facilitation effects due to associative relations were found. The absence of effects at T0 could indicate that the self-paced reading is somewhat ahead of the actual semantic processing of the words and their semantic relations. It is therefore that the spillover region was included in the analyses as well. We did not find semantic priming effects for associatively related word pairs, which appears counter-intuitive at first sight. Not only was the associative boost, i.e. additional facilitation for associated versus non-associated word pairs (Lucas, 2000), not borne out in our data, we did not find a significant effect for associated word pairs altogether. Other studies of semantic priming at the sentence level have typically used associated word pairs, with mixed results. While there are studies that report associative boost effects (e.g. Myers, Cook, Kambe, Mason, & O'Brien, 2000) or facilitation

of associated pairs but only a marginal effect for non-associated pairs (Carroll & Slowiaczek, 1986), others found no evidence for any associative facilitation (Morris, 1994; Traxler, Foss, Seely, Kaup, & Morris, 2000). Some researchers suggest that associative priming within sentences is highly susceptible to variation in the previous context (Camblin, Gordon, & Swaab, 2007; Hess, Foss, & Carroll, 1995) and may only be observed in early measures of reading, such as initial eye landing spots and first-pass reading times (Camblin et al., 2007; Lavigne, Vitu, & d'Ydewalle, 2000). Within the framework of Van den Broek's Landscape model one could state that semantic relations between words are just one of the sources for spreading of activation during sentence reading, activation that may not always be notable, or determinative for good comprehension. Other measures, such as eye-tracking, that are more sensitive to early processes, or more neurologically-oriented ones could potentially better detect the effects of associative semantic priming.

Because semantic priming studies at the sentence level often equate association to semantic relation or compare associated to non-associated pairs, there is less evidence for pure priming of non-associated word pairs during sentence reading. Carroll and Slowiaczek (1986) found a marginal effect at their non-associatively related category coordinate targets, but did not examine the post-target regions. Our two effects of non-associated priming suggest that it may affect later measures especially, with a weaker general effect of both context-dependent and context-independent pairs at S1, and a stronger effect of context-independent pairs only at S2. Facilitation for context-dependent items was thus only observed in conjunction with the context-independent set. The effect of non-associative, context-independently related word pairs we found at S2 is in accordance with previous findings from single word priming studies in the same target population: Cremer (2013) and Spätgens and Schoonen (2018) found such facilitation in the written and auditory modality, respectively.

4.2. Semantic priming and reading comprehension

Following the findings by Nation and Snowling (1999), we hypothesized that especially non-associated, context-independent priming would be predictive of reading comprehension scores. However, although we found delayed facilitation of exactly this semantic type at S2 (Table 8), it was not a significant predictor of the reading comprehension scores (Table 11). The marginal effect we did find, a positive relation between non-associated priming at S1 and reading comprehension (Table 10), is tentative at best, and did not significantly improve the reading comprehension model. Clearly, the data do not warrant a strong interpretation of this effect.

It appears that although we expected to approximate the process of semantic activation during reading more closely by studying word pairs in a sentence context, this was not sufficient to identify a clear relation to reading comprehension on an individual differences level. This suggests that the effects found in clearly distinct groups of readers, as studied by Nation and Snowling (1999) and Bonnotte and Casalis (2010) are not borne out in individual differences in a sample of normal readers, as was also concluded in Cremer (2013) and Spätgens and Schoonen (2018).

If semantic activation of related words is used during reading comprehension, as is for example posited in the Landscape model (Van

⁶ For reasons of brevity, the base models are not reported on in full here.

Table 11Fixed effects estimates for predictors of reading comprehension scores, including non-associated, context-independent priming at the S2 position ($N = 137$).

	Estimate (se)		<i>t</i>	<i>p</i>
Intercept	11.63	(2.18)	5.3	0.000***
Vocabulary size	0.37	(0.09)	4.0	0.000***
Context-independent semantic knowledge	0.26	(0.09)	3.0	0.003**
Word decoding	0.01	(0.02)	0.2	0.819
S2 non-associated, context-independent priming	4.45	(3.32)	1.3	0.180
Language group	0.27	(0.70)	0.4	0.699
S2 non-associated, context-independent priming $\times$ language group	-6.30	(6.57)	-1.0	0.338

** $p < 0.01$.*** $p < 0.001$.

den Broek et al., 1999, 2005), an absence of individual difference effects on comprehension is unexpected. However, priming effects are very subtle and subject to the influence of a wide range of experiment-internal and -external factors. Therefore, it may be the case that even if semantic priming contributes directly to extra activation and faster connections between concepts needed to establish coherence while reading, it may be that this spreading activation not necessarily leads to better (offline) comprehension of texts. To capture such subtle relations, more sensitive and early measures of reading processes such as those obtained in eye tracking may be a solution. However, Stolz, Besner, and Carr (2005) and Yap, Hutchison, and Tan (2016) showed that although group-level priming effects are generally robust, individuals do not show consistent facilitation across sessions. Translating individual differences from such a measure to individual differences in other skills may therefore be complicated altogether, especially when priming effects are obtained from a single session.

In sum, the results do confirm the principal role of vocabulary size and context-independent semantic knowledge for reading comprehension, as is claimed by the Lexical Quality Hypothesis (Perfetti, 2007). However, because no significant effects of semantic priming on the reading comprehension scores were found from this data set,⁷ we cannot assert that semantic priming makes a major contribution to reading comprehension. This suggests that even if future studies are able to identify significant effects of semantic priming on reading comprehension on an individual reader's level, the influence of this factor will be much smaller than that of established vocabulary measures such as size and depth.

4.3. Differences between monolinguals and bilinguals

Most of the test performances of the monolingual and bilingual children were comparable. It seems that these (sequential) bilingual children, who were educated in the same school setting, have caught up with their monolingual peers. This is also supported by the lack of heterogeneity in our sample (see the similar standard deviations in comparison to the monolingual group in Table 3) that is often presumed in groups of bilingual or second language learners. The only significant differences observed between monolinguals and bilinguals were lower scores on declarative context-independent semantic knowledge as was measured with a structured word association task (WAT), and faster reading times for bilinguals at most target and spillover positions in the self-paced reading task, independent of relatedness or semantic type. The lower performance on the WAT has been identified in multiple previous studies (Cremer, 2013; Schoonen & Verhallen, 2008), but may be all the more interesting in the present sample given that differences in vocabulary size and reading comprehension were not observed. Limitations in declarative context-independent knowledge may thus

persist in bilinguals even when their vocabulary size is on par with that of their monolingual peers. Apparently then, building the number of vocabulary items does not necessarily coincide with building the quality or depth of knowledge on these items, which is in line with studies that have shown that vocabulary size is a measure separate from other dimensions of vocabulary knowledge such as depth (see Schmitt, 2014, for an overview). However, these limitations in declarative context-independent knowledge did not translate to differences in semantic priming between monolinguals and bilinguals. This would appear counter-intuitive at first sight, but it was also the case in the sample studied by Cremer (2013). Working with the concept of context-independency in a declarative task such as the WAT may be more difficult for bilinguals, even if some level of automatic activation of this knowledge is already in place.

The faster reading times for bilinguals in the self-paced reading experiment may point to a subtle advantage in word decoding speed, which would be in accordance with small advantages in decoding speed for that have been found in other studies (Melby-Lervåg & Lervåg, 2014; Raudszus, Segers, & Verhoeven, 2018). However, it may also be a performance difference specific to this task, since there were no group differences in the word decoding task itself.

Most notable however, is the absence of significant group differences on most measures, including semantic priming, in line with Spätgens and Schoonen (2018). Our selection of schools in neighborhoods with low to average SES and the fact that the bilingual children mostly spoke Dutch at home in addition to another language, likely plays a role in these results. Recent national assessment reports on the Dutch school population (e.g. Kuhlemeier et al., 2014) also show that monolinguals and bilinguals in the current Dutch context show similar language performance when SES is controlled for. The role bilinguals' first language may play was not part of this study, and we therefore did not collect information about L1 proficiency. Given the homogeneity in test scores of the bilingual group, their similar school careers and the fact that they became literate in Dutch, we do not expect that differences in L1 proficiency have affected the findings reported here, however, future research needs to corroborate this expectation.

4.4. Limitations and directions for future research

There are a few limitations to the self-paced reading experiment and the other tasks used in our study, which may be improved upon in future studies. Regarding the self-paced reading experiment, testing in a highly controlled lab session would have been preferred, because priming effects are concerned with such minute differences in behavioral measures. In our target population however, this was not a viable option, and testing in the school setting may have affected the potential of our experiment to capture these subtle effects. However, we did find priming (condition) effects, but generally speaking smaller effects than those reported by Nation and Snowling (1999) or Bonnotte and Casalis (2010). Given the general instability of individual priming effects discussed earlier, it turns out to be difficult to relate individual priming effects to reading comprehension scores.

⁷ It was also not the case that a potential effect of the priming measures was already captured by the vocabulary size influence, because bivariate correlations between the priming scores and reading comprehension were non-significant.

Future studies may also be able to improve on some of the features that affected the strength of our experiment. Our attempt to study the facilitative effect of a 2×2 contrast of semantic relations did mean that the number of semantic pairs that satisfied all necessary criteria was somewhat limited within each set. With larger sets of related and unrelated pairs, additional priming effects or a clearer relation to reading comprehension might have been found. Students' sensitivity to subtle semantic priming effects might be better captured with more sensitive procedures like eye-tracking or neuro-imaging (e.g., N400).

Finally, the use of highly frequent words guaranteed that the words were known to the students and enabled us to disentangle priming effects from potential vocabulary size effects. However, because these words were so familiar, individual differences in activation between them may not have been sufficiently large in this age group to be predictive of reading comprehension differences at fifth-grade levels. In this respect it would be interesting to be able to include words that are just known to the students and that may show larger individual differences in semantic priming.

With regard to the other tasks, higher reliability scores would have been desirable. In addition, as Swart et al. (2017) have pointed out, reading comprehension is a complex process, which may best be assessed by means of multiple tasks to tap all its facets. Furthermore, we have to bear in mind that the reading comprehension test was offline and did not put any time pressure on the students, and that the words in the self-paced reading task used to collect semantic priming data were unrelated to the reading texts, which implies that –in the second part of our study (Section 3.3), we were aiming at a general feature of the students' mental lexicon, in relation to their general reading proficiency.

Appendix A. Related word pairs

Table A.1
Word pairs including English translation.

Context-independent pairs								
Non-associated					Associated			
chocola	drop	chocolate	liquorice	*	auto	vervoer	car	transport
eend	gans	duck	goose	*	banaan	fruit	banana	fruit
egel	mol	hedgehog	mole	*	bruid	vrouw	bride	woman
fiets	brommer	bike	moped	*	gitaar	instrument	guitar	instrument
hamer	boor	hammer	drill		hamster	huisdier	hamster	pet
hond	wolf	dog	wolf	*	kaas	melk	cheese	milk
jam	honing	marmalade	honey		komkommer	groente	cucumber	vegetable
jas	muts	coat	cap		koning	baas	king	boss
kat	tijger	cat	tiger	*	krekel	insect	cricket	insect
regen	sneeuw	rain	snow	*	oceaan	zee	ocean	sea
trompet	fluit	trumpet	flute	*	pop	speelgoed	doll	toy
trui	broek	sweater	pants	*	rok	kleding	skirt	clothing
zebra	ezel	zebra	donkey	*	uil	vogel	owl	bird
zwaan	meeuw	swan	sea gull	*	vlinder	rups	butterfly	caterpillar
Context-dependent pairs								
Non-associated					Associated			
bakkerij	klant	bakery	customer	*	baby	luier	baby	diaper
bos	eekhoorn	forest	squirrel	*	boer	vee	farmer	cattle
chauffeur	stuur	driver	steering wheel	*	camping	tent	camp site	tent
dief	tas	thief	bag	*	heks	hoed	witch	hat
dokter	pleister	doctor	band-aid	*	indiaan	veren	indian	feathers
juf	lokaal	teacher (female)	classroom	*	jager	geweer	hunter	rifle
kapitein	haven	captain	harbor	*	kast	spullen	closet	stuff
kapper	borstel	hair dresser	brush	*	kasteel	ridder	castle	knight
kok	oven	chef	oven	*	markt	kleren	market	clothes
prinses	koets	princess	carriage	*	piloot	vliegtuig	pilot	air plane
station	conducteur	train station	conductor	*	schilder	verf	painter	paint
tuin	mus	garden	sparrow	*	stal	hooi	stable	hay
visser	rivier	fisherman	river	*	toerist	vakantie	tourist	holiday

(continued on next page)

4.5. Concluding remarks

Despite these limitations, there is clear consistency between our findings and previous research, in which non-associated context-independent priming was also observed, but not found to be predictive of individual differences in reading comprehension. This suggests that either relating a relatively unstable measure such as priming to a complex skill such as reading comprehension is not feasible on an individual level, or that an individual's degree of semantic activation, even when captured during reading, simply has little correspondence to his or her reading comprehension performance. When we consider the study of component skills of vocabulary knowledge as a possible starting point to guide intervention efforts supporting reading comprehension, individual differences in semantic priming do not seem to be a major source of reading difficulties in a common population of readers.

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Table A.1 (continued)

Context-dependent pairs								
Non-associated					Associated			
woestijn	slang	desert	snake	*	winkel	geld	store	money

The non-associated pairs with * were also used in Spätgens and Schoonen (2018).

Appendix B. Length and frequency for primes and targets

Table B.1

Length (in letters) and frequency information for primes and targets.

	Context-independent, non-associated		Context-independent, associated		Context-dependent, non-associated		Context-dependent, associated	
	M	SD	M	SD	M	SD	M	SD
Length related primes	4.6	1.3	5.4	1.8	5.9	2.1	5.6	1.5
Length control primes	5.0	1.5	5.2	1.1	6.1	2.0	5.3	1.4
Frequency related primes	270.9	297.6	173.6	242.6	281.6	325.7	181.8	128.5
Frequency control primes	250.0	292.5	165.8	185.4	272.6	311.1	181.3	131.2
Length targets	4.8	1.1	6.0	2.0	5.7	1.9	5.4	1.7
Frequency targets	143.1	96.2	225.5	255.0	102.4	91.8	191.3	164.4

Frequency data from Schrooten and Vermeer (1994)

Appendix C. Test sentences

Table C.1

Experimental sentence pairs (prime-target pairs in boldface, related items always followed by unrelated items) by semantic type and plausibility.

Context-independent, non-associated	
Plausible items	Implausible items
Kevin mocht chocola en drop meenemen toen we op kamp gingen. Kevin mocht limonade en drop meenemen toen we op kamp gingen. Ik kan alleen de hamer of de boor meenemen want allebei wordt te veel. Ik kan alleen de plank of de boor meenemen want allebei wordt te veel. Ik zag bij die egel een mol zitten en dat is toch wel bijzonder. Ik zag bij die struik een mol zitten en dat is toch wel bijzonder. Amir speelt trompet en fluit en kan ook heel goed zingen. Amir speelt voetbal en fluit en kan ook heel goed zingen. Het nieuws ging over de regen en de sneeuw waar iedereen last van heeft op de weg. Het nieuws ging over de drukke en de sneeuw waar iedereen last van heeft op de weg. Het was een film over een hond en kan ook heel goed zingen. Het was een film over een meisje en een wolf die hele goede vrienden werden. Het plaatje met de eend en de gans vind ik niet zo mooi. Het plaatje met de hut en de gans vind ik niet zo mooi.	Dina tekent een kat naast de tijger op de kleine zachte kauwgom. Dina tekent een zon naast de tijger op de kleine zachte kauwgom. Onderweg zag ik een fiets bij een brommer langs de kant staan printen. Onderweg zag ik een paard bij een brommer langs de kant staan printen. Op het schilderij zag je een zwaan naast een meeuw met grote vleugels scholen eten. Op het schilderij zag je een hek naast een meeuw met grote vleugels scholen eten. Is hij nou zijn trui en zijn broek weer vergeten mee te bakken? Is hij nou zijn deken en zijn broek weer vergeten mee te bakken? Het verhaal ging over een zebra en een ezel die samen avonturen in de vloer wassen. Het verhaal ging over een rover en een ezel die samen avonturen in de vloer wassen. Hij pakte zijn jas en zijn muts om mee te kunnen hoesten in het krijtje. Hij pakte zijn bril en zijn muts om mee te kunnen hoesten in het krijtje. Zet je de jam en de honing ook terug in de vulkaan? Zet je de olie en de honing ook terug in de vulkaan?
Context-dependent, non-associated	
Plausible items	Implausible items
Als je in dit bos een eekhoorn zou vinden, is dat niet gek.* Als je in dit schip een eekhoorn zou vinden, is dat wel gek.* Mijn kapper is altijd haar borstel kwijt als ze hem nodig heeft. Mijn dochter is altijd haar borstel kwijt als ze hem nodig heeft. Hij had in de woestijn een slang gezien waardoor hij erg was geschrokken. Hij had in de speeltuin een slang gezien waardoor hij erg was geschrokken. Toen zijn juf het lokaal in liep was het een grote bende. Toen zijn oma het lokaal in liep was het een grote bende. Op het station zocht de conducteur naar de man die hem had geroepen. Op het kantoor zocht de conducteur naar de man die hem had geroepen. De oude kapitein kijkt naar de haven terwijl de grote boot wegvaart. De oude mevrouw kijkt naar de haven terwijl de grote boot wegvaart. Mijn kok heeft alvast de oven aangezet voor zijn speciale koekjes. Mijn neef heeft alvast de oven aangezet voor zijn speciale koekjes.	In de bakkerij heeft de klant een praatje gemaakt met de snoeischaar. In de woonkamer heeft de klant een praatje gemaakt met de snoeischaar. De jonge visser kijkt naar de rivier terwijl de maan gras maait. De jonge keizer kijkt naar de rivier terwijl de maan gras maait. De chauffeur pakte het stuur wat steviger vast in de blauwe sla. De directeur pakte het stuur wat steviger vast in de blauwe sla. In een grote tuin kan een mus heel ver vliegen met tafels. In een grote stad kan een mus heel ver vliegen met tafels. De aardige dokter pakte een pleister uit de grote televisie met lepels. De aardige dame pakte een pleister uit de grote televisie met lepels. Ik wil niet dat die dief mijn tas mee kan nemen uit de pindakaas. Ik wil niet dat die kerel mijn tas mee kan nemen uit de pindakaas. De prinses heeft de koets al horen aankomen op de kroket. De soldaat heeft de koets al horen aankomen op de kroket.
Context-independent, associated	
Plausible items	Implausible items
Cas houdt van banaan en rood fruit in zijn yoghurt bij het ontbijt. Cas houdt van suiker en rood fruit in zijn yoghurt bij het ontbijt.	Milan houdt van de oceaan en de zee en wandelen in de scherpe papiertjes. Milan houdt van de lente en de zee en wandelen in de scherpe papiertjes.

(continued on next page)

Table C.1 (continued)

Context-independent, associated	
Plausible items	Implausible items
Zara wil een gitaar en een instrument met van die belletjes, een tamboerijn. Zara wil een puzzel en een instrument met van die belletjes, een tamboerijn. Op een krekel lijkt dit insect wel een beetje, vind ik. Op een schelp lijkt dit insect wel een beetje, vind ik. Emma maakt een tekening van een vlinder en een rups met bloemen en gras erbij. Emma maakt een tekening van een vijver en een rups met bloemen en gras erbij. De gemene koning is de baas van het hele land in het sprookje. De gemene leeuw is de baas van het hele land in het sprookje. Die oude pop als speelgoed gebruiken is geen goed idee, leg hem maar terug. Die oude foto als speelgoed gebruiken is geen goed idee, leg hem maar terug. Mamma wil nog een rok en warme kleding voor Maaïke kopen deze week. Mamma wil nog een vaas en warme kleding voor Maaïke kopen deze week.	Daan zag de bruid bij de vrouw in de kleine kruimel lopen. Daan zag de wesp bij de vrouw in de kleine kruimel lopen. Door de auto ging het vervoer van het grote oerwoud naar de tranen wel goed. Door de lucht ging het vervoer van het grote oerwoud naar de tranen wel goed. Ik wil geen hamster als huisdier maar een grote zonnige stoetegel. Ik wil geen garnaal als huisdier maar een grote zonnige stoetegel. De uil en de kleine vogel zitten stil in de nietmachine. De zoon en de kleine vogel zitten stil in de nietmachine. Pappa moest nog kaas en melk halen in de groene bijl. Pappa moest nog zeep en melk halen in de groene bijl. Neem je nog een komkommer en groente uit de diepvries mee van de verwarming? Neem je nog een pudding en groente uit de diepvries mee van de verwarming?
Context-dependent, associated	
Plausible items	Implausible items
Gelukkig heeft de oude boer nog vee en ook veel land waar hij op werkt. Gelukkig heeft de oude heer nog vee en ook veel land waar hij op werkt. Ik had in de winkel mijn geld laten liggen, echt stom van me. Ik had in de trein mijn geld laten liggen, echt stom van me. Bij een kasteel was de ridder gestopt om uit te rusten. Bij een heuvel was de ridder gestopt om uit te rusten. De indiaan heeft de veren gevonden in het hoge gras. De knecht heeft de veren gevonden in het hoge gras. In het verhaal raakte de heks haar hoed kwijt in het donkere woud. In het verhaal raakte de zus haar hoed kwijt in het donkere woud. De aardige schilder heeft de verf nog even uit zijn busje gehaald. De aardige buurman heeft de verf nog even uit zijn busje gehaald. De bange jager liet het geweer per ongeluk op de grond vallen. De bange prins liet het geweer per ongeluk op de grond vallen.	Er stond een piloot naast het vliegtuig op de landingsbaan van trommels. Er stond een ladder naast het vliegtuig op de landingsbaan van trommels. Voor de camping stond een tent met veel groene slagroom en sleutels. Voor de garage stond een tent met veel groene slagroom en sleutels. Zijn kleine baby wilde geen luier aan omdat hij de straat moest typen. Zijn kleine broer wilde geen luier aan omdat hij de straat moest typen. Op de markt zijn mijn kleren helemaal vies geworden van het nadenken. Op de kermis zijn mijn kleren helemaal vies geworden van het nadenken. Bij de stal lag het hooi met alle mooie telefoons erdoorheen. Bij de put lag het hooi met alle mooie telefoons erdoorheen. De toerist vond zijn vakantie heel erg leuk en eetbaar. De postbode vond zijn vakantie heel erg leuk en eetbaar. Daar achter de kast vind je de spullen waarmee je de beker kunt oefenen. Daar achter de muur vind je de spullen waarmee je de beker kunt oefenen.

* This pair differs in the penultimate word to make sure both sentences are plausible, but since this position is outside of the spillover region, this difference does not affect our measurements of interest.

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